Test 10 Innovator
Math Module 1

1. Scott works at a job for four hours. He spent $\$18$ of his earnings for lunch, and has $\$42$ remaining. How many dollars did he earn per hour at the job?

PREV. QUESTION | NEXT QUESTION

× A 6.00

× B 10.50

× C 15.00

× D 60.00

2. Line A is perpendicular to line B. The equation for line A is $y = \frac{3}{7}x - 5$. What is the slope of line B?

PREV. QUESTION | NEXT QUESTION

× A -5

× B $-\frac{7}{3}$

× C $\frac{3}{7}$

× D $\frac{7}{3}$

3. $$2x - 3y = -15$$
$$3x + 3y = 30$$

What is the solution (x, y) to the given system of equations?

PREV. QUESTION | NEXT QUESTION

× A $(-3, 3)$

× B $(3, 7)$

× C $(6, 9)$

× D $(9, 1)$

4. Desmond wants to have at least $\$1{,}000$ in his savings account by the end of the summer. He starts the summer with $\$175$, and he earns $\$100$ a week at his job. Which of the following inequalities represents the possible number of weeks, x, he needs to work this summer?

PREV. QUESTION | NEXT QUESTION

× A $1{,}000 \leq 100x + 175$

× B $1{,}000 \geq 100x + 175$

× C $1{,}000 \leq 175x + 100$

× D $1{,}000 \geq 175x + 100$

5. $$x = 5$$
$$2x + y = 13$$

The graphs of the equations in the given system of equations intersect at the coordinate point (x, y) when graphed in the xy–plane. What is the product of x and y?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

6. $$10x + 7x^2 - 9x$$

Which of the following is equivalent to the given expression?

PREV. QUESTION | NEXT QUESTION

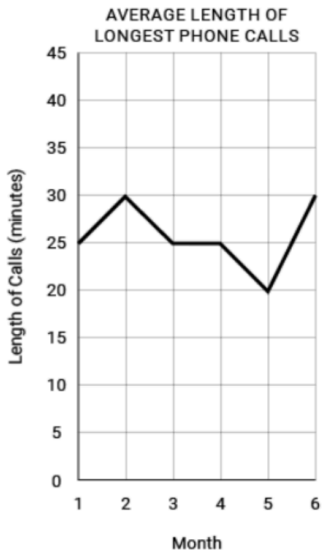
× A $7x^2 + x$

× B $7x^2 + 1$

× C $7x^2 + 19x$

× D $8x^2$

7.



A phone company collects data on a random sample of 300 customers for six months. They determined the longest phone call each customer made, and then found the average time of the longest phone call for each month. The results are shown in the line graph. Which of the following can be inferred about all of the phone company's customers?

PREV. QUESTION | NEXT QUESTION

× A The longest call of any customer is 30 minutes.

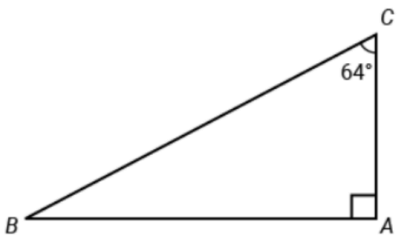
× B The average time for the longest phone call in the fifth month is 20.

× C All customers made at least one phone call each month.

× D The longest phone call anyone made during month 3 was 25 minutes.

COME BACK LATER

8.



Note: Figure not drawn to scale.

Right triangle ABC is shown. What is the measure of angle B, in degrees?

PREV. QUESTION | NEXT QUESTION

× A 26

× B 32

× C 64

× D 116

COME BACK LATER

9. $(-x^4 + 9) - (-7x^4 - 17)$

The expression shown is equivalent to $kx^4 + 26$, where k is an integer. What is the value of k?

PREV. QUESTION | NEXT QUESTION

_____

COME BACK LATER

10. Sphere A has a diameter of 18 centimeters. Sphere B has a diameter equal to the radius of sphere A. What is the volume of sphere B, in cubic centimeters?

PREV. QUESTION | NEXT QUESTION

× A 20.25π

× B 121.5π

× C 288π

× D 972π

COME BACK LATER

11. $-11x + 4 = -11x - 4$

How many solutions does the given equation have?

PREV. QUESTION | NEXT QUESTION

× A Exactly one

× B Exactly two

× C Infinitely many

× D Zero

12. Line k has a slope of -3 and passes through the point $(-0.5, 4)$ when graphed in the xy-plane. Which of the following is the equation of line k?

PREV. QUESTION | NEXT QUESTION

× A $y = -3x + 2.5$

× B $y = -3x + 4$

× C $y = -3x + 5.5$

× D $y = -3x + 11$

13. $$3(x + 7) \leq 5(x - 1)$$

Which of the following describes the solution set to the given inequality?

PREV. QUESTION | NEXT QUESTION

× A $x \leq 4$

× B $x \geq 4$

× C $x \leq 13$

× D $x \geq 13$

14. $$8x^2 - 6x - 27 = 0$$

What is the negative solution to the given equation?

PREV. QUESTION | NEXT QUESTION

× A $-\frac{27}{8}$

× B $-\frac{9}{4}$

× C $-\frac{3}{2}$

× D $-\frac{2}{3}$

COME BACK LATER

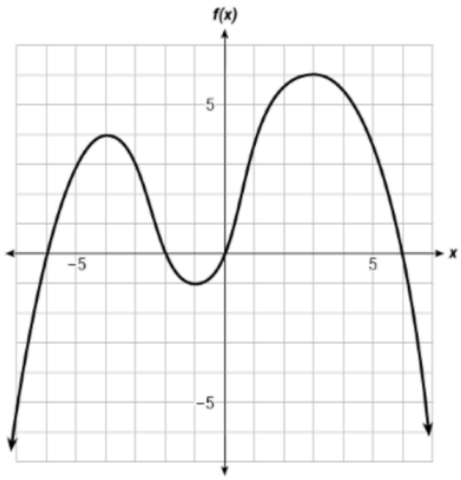
15. $$a(x+4)=b(x-1)+1$$

If the solution to the given equation is $x=4$, which of the following could be a value of a and b?

PREV. QUESTION | NEXT QUESTION

- A $a=8$ and $b=-3$
- B $a=7$ and $b=6$
- C $a=4$ and $b=-5$
- D $a=-4$ and $b=-11$

16.



The function $f(x)$ is graphed on the coordinate plane, where the function f is defined by $f(x)=ax^4+bx^3+cx^2+dx+e$ and a, b, c, d, and e are constants. For how many values of x does $f(x)=0$?

PREV. QUESTION | NEXT QUESTION

- A Zero
- B One
- C Two
- D Four

COME BACK LATER

17. $$y=x^2-10x+b$$
$$y=4.25$$

In the given system of equations, b is a constant. The system has exactly one solution. What is the value of b?

PREV. QUESTION | NEXT QUESTION

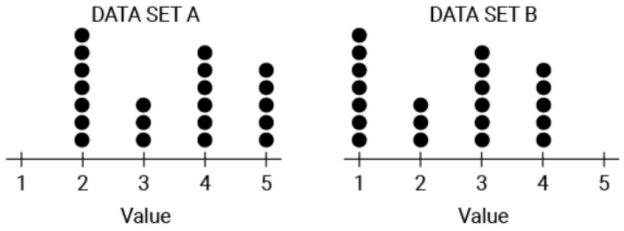
COME BACK LATER

18. The hypotenuse of a right isosceles triangle is $j\sqrt{k}$, where j and k are rational numbers. The leg of the right isosceles triangle is the same length as the hypotenuse of a right triangle with angles measuring 30°, 60°, and 90°. If the side opposite the 60° angle is $3.25\sqrt{3}$, what is the value of $j + k$?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

19.

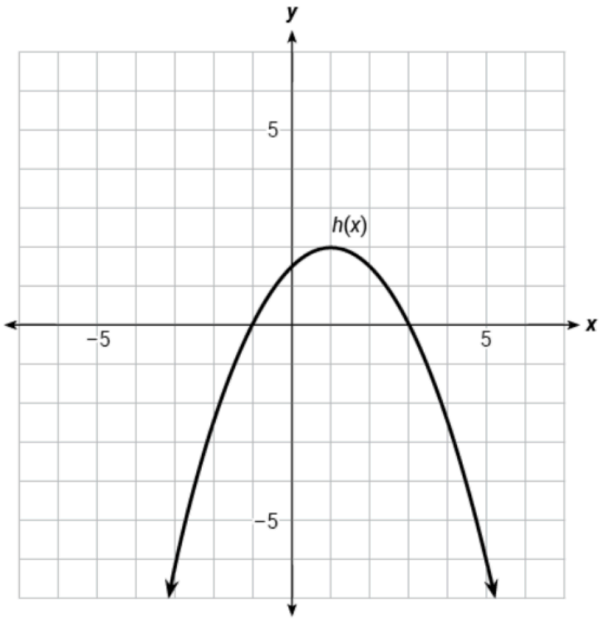


The dot plot shows the distribution of values in two data sets. Which of the following is true about the standard deviations of two data sets?

PREV. QUESTION | NEXT QUESTION

- A The standard deviation is greater in set A than in set B.
- B The standard deviation is greater in set B than in set A.
- C The standard deviation is the same in set A and set B.
- D There is not enough information to determine the relationship between the standard deviations of the two data sets.

20.



The quadratic function $h(x) = ax^2 + bx + c$ is graphed on the xy–plane. Which of the following is the equation for $h(x)$?

PREV. QUESTION | NEXT QUESTION

A $h(x) = -x^2 + 2x + 3$

B $h(x) = -\frac{1}{2}x^2 - x + \frac{3}{2}$

C $h(x) = -\frac{1}{2}x^2 + x + \frac{3}{2}$

D $h(x) = \frac{1}{2}x^2 - x - \frac{3}{2}$

COME BACK LATER

21.

x	f(x)
−20	25
−17	19
−14	13
−11	7

The table shows various input and output values for the linear function $f(x)$. If $f(x)$ is graphed on the xy–plane, which coordinate is the x–intercept of the graph?

PREV. QUESTION | NEXT QUESTION

A $(-15, 0)$

B $(-\frac{15}{2}, 0)$

C $(\frac{15}{2}, 0)$

D $(15, 0)$

COME BACK LATER

22. The number a is 80% greater than the positive number b. The number c is 60% less than a. The value of c is how many times the value of b?

PREV. QUESTION

× A 0.32

× B 0.72

× C 1.08

× D 4.50

MATH MODULE 2B (HARDER)

1. The center point of circle Q when graphed in the xy-plane is $(4, 6)$. The coordinate $(4, 11)$ is on the circumference of the circle. What is the diameter of circle Q?

NEXT QUESTION

× A 4

× B 5

× C 10

× D 12

2. $$3x - x(7 - 8) + 12 = 4x - (6x + 18)$$

In the given equation, what is the value of x?

PREV. QUESTION NEXT QUESTION

× A -5

× B -1

× C 1

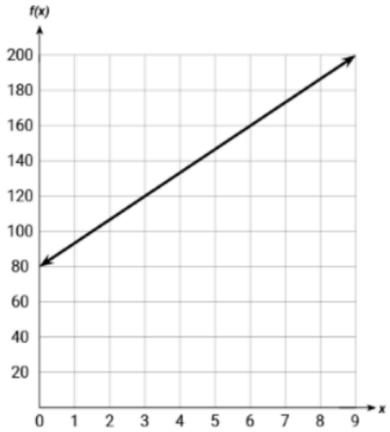
× D 5

3. The equation for line p is $5x - 8y = 40$. Line q is perpendicular to line p and passes through the point $(10, 0)$. What is the equation for line q?

PREV. QUESTION | NEXT QUESTION

- A $y = -\frac{8}{5}x + 16$
- B $y = -\frac{8}{5}x + 10$
- C $y = \frac{5}{8}x + 16$
- D $y = \frac{5}{8}x + 10$

4.



The graph shows how much money $f(x)$ Maya saves over x months. What is the best interpretation of the slope of the function $f(x)$?

PREV. QUESTION | NEXT QUESTION

- A Maya saves $\$20$ each month.
- B Maya saves $\$40$ every 3 months.
- C Maya saves $\$80$ each month.
- D Maya saves $\$120$ in a year.

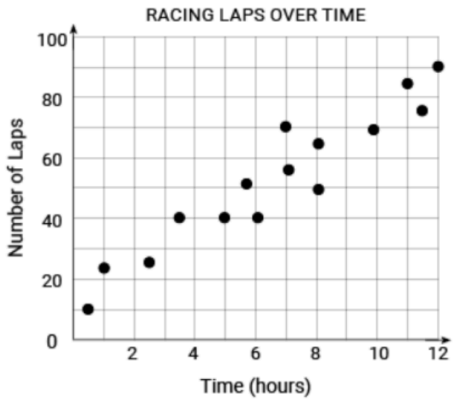
COME BACK LATER

5. Yuv is purchasing shoes that originally cost $\$200$. The store is having a 20% off sale. Yuv has a coupon for an additional 10% off that is applied after the store discount. He can determine the price of the shoes with both discounts by calculating $(200)(p)$. What is the value of p?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

6.



A number of racecar racers continuously go around a track. The scatterplot shows the distribution of how many laps the racers could complete after a certain time. If y is the number of laps and x is the time in hours, which of the following is a line that most closely fits the data?

PREV. QUESTION | NEXT QUESTION

A $y = 0.7x + 15$

B $y = 2x + 30$

C $y = 7x + 10$

D $y = 10x$

COME BACK LATER

7. The diagonal of a rectangle is 26 meters. The width of the rectangle is 10 meters. What is the area of the rectangle, in square meters?

PREV. QUESTION | NEXT QUESTION

- A 120
- B 130
- C 240
- D 260

8. Martha wants to know what dances are popular on social media platforms she uses by conducting a survey. Which group of people should Martha survey to get the most accurate data?

PREV. QUESTION | NEXT QUESTION

- A A random sample of followers on a random social media platform
- B A random sample of users of all the social media platforms that Martha uses
- C A random sample of users who follow Martha across all of her social media platforms
- D A random sample of Martha's followers who are also dancers

9. $x^2 - 6x + 7 = 0$

What values of x satisfy the given equation?

PREV. QUESTION | NEXT QUESTION

A $x = 1$ and $x = 7$

B $x = -1$ and $x = -7$

C $x = 3 + \sqrt{2}$ and $x = 3 - \sqrt{2}$

D $x = -3 + \sqrt{2}$ and $x = -3 - \sqrt{2}$

10. $\frac{3x+2}{ax+5} + 2x - 3 = \frac{8x^2+x-13}{ax+5}$

In the expression above, $x \neq \frac{-5}{a}$ and a is a non-zero constant. What is the value of a?

PREV. QUESTION | NEXT QUESTION

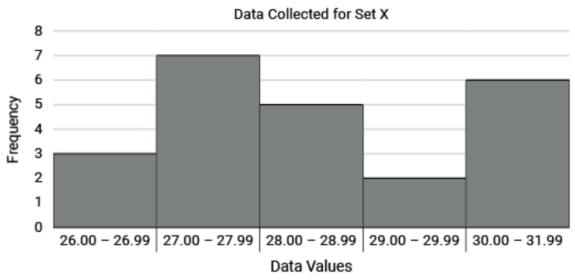
A -4

B $-\frac{1}{3}$

C 2

D 4

11.



The histogram shows the data collected from set X. Data set Y takes all the values from set X and subtracts 17.5 to every value. Which of the following correctly compares the median and the range between set X and set Y?

PREV. QUESTION | NEXT QUESTION

A. The median for set X is greater than the median for set Y and the range for set X is greater than the range of set Y.

B. The median for set X is greater than the median for set Y and the range for set X is equal to the range of set Y.

C. The median for set X is equal to the median for set Y and the range for set X is greater than the range of set Y.

D. The median for set X is equal to the median for set Y and the range for set X is equal to the range of set Y.

12.

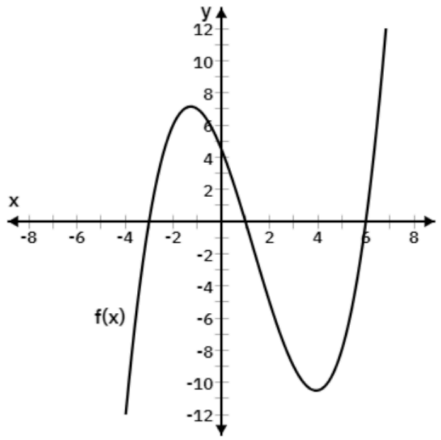
$$4|12-2x|+12|6-x|=140$$

What is the positive solution to the given equation?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

13.



The graph of $y = f(x)$ is graphed on the xy–plane. The equation of $f(x)$ is represented by $f(x) = a(x + m)(x + n)(x + o)$, where a, m, n, and o are constants. If the graph of $f(x)$ is moved one unit to the left, which of the following is true?

I. The value of a decreases by 1
II. The product of m, n, and o is 0
III. The y–intercept of the translated graph is at the origin

PREV. QUESTION | NEXT QUESTION

A I only

B I, II, and III

C II only

D II and III

COME BACK LATER

14. Function g is defined by the equation $g(x) = 8(\frac{1}{4})^x$. If $h(x) = g(x + 2)$, which of the following is the equation for $h(x)$?

PREV. QUESTION | NEXT QUESTION

A $h(x) = \frac{1}{2}(\frac{1}{4})^x$

B $h(x) = 4(\frac{1}{4})^x$

C $h(x) = 8(\frac{1}{4})^x + 2$

D $h(x) = 16(\frac{1}{2})^x$

15.

x	$f(x)$
−5	−90
−3	0
0	0
2	−20
4	0
5	40

The table above shows several values of the function f graphed on the xy–coordinate plane. The equation of f is defined by $f(x) = (x + a)(x + b)(x + c)$, where a, b, and c are constants. Which of the following is true?

I. $a + b + c < 0$
II. $f(-2) < 0$
III. $f(-6) < f(6)$

PREV. QUESTION NEXT QUESTION

A I only

B I and III

C I, II and III

D III only

COME BACK LATER

16.

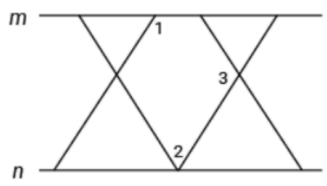
$$0.5(1.5x - 0.75) = 3(-\tfrac{x}{6} + \tfrac{1}{12})$$

What is the solution to the given equation?

PREV. QUESTION NEXT QUESTION

COME BACK LATER

17.



Note: Figure not drawn to scale

In the image shown, line m is parallel to line n, all four triangles formed by the intersecting lines are isosceles triangles, and the two smaller triangles are congruent. If the measure of angle 3 is 104°, what is the sum of angle 1 and angle 2, in degrees?

PREV. QUESTION | NEXT QUESTION

- A 128
- B 204
- C 208
- D 256

COME BACK LATER

18. Let t represent the number of tulips and r represent the number of roses in a bouquet. Each tulip costs $\$3.50$ and each rose costs $\$5.25$. Flora wants to buy a bouquet for at most $\$50$ and have at least three times as many tulips as roses in the bouquet. Which system of inequalities best represents this scenario?

PREV. QUESTION | NEXT QUESTION

- A $3.50t + 5.25r \leq 50$, $t \leq 3r$
- B $3.50t + 5.25r \leq 50$, $3t \leq r$
- C $3.50t + 5.25r \leq 50$, $t \geq 3r$
- D $3.50t + 5.25r \leq 50$, $3t \geq r$

19. The linear function $f(x)$ is defined such that $f(2) = 13.5$ and $f(14) = 22.5$. If $f(x+1) = 18.75$, what is the value of x?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

20. A quadratic equation with a leading coefficient of 1 intersects the x–axis at the points $(a, 0)$ and $(b, 0)$. What is the value of the y–coordinate of the vertex?

PREV. QUESTION | NEXT QUESTION

A $-\frac{1}{4}(b-3a)(b-a)$

B $-\frac{1}{4}(a-b)^2$

C $\frac{1}{4}(a-b)^2$

D $\frac{1}{4}(b-3a)(b-a)$

21. $$f(t) = 155(1.14)^{\frac{4}{3}t}$$

In the given function, the growth in revenue at a certain company is modeled, where t represents years. According to the model, how many months does it take for the revenue to increase by 14%?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

22. Two similar cylinders, Cylinder A and Cylinder B, have volumes, in cubic meters, of 100π and 800π respectively. If the radius of Cylinder A is equal to 2 meters, what is the total surface area of Cylinder B in square meters? (Surface area of a cylinder $= 2\pi r(r + h)$)

PREV. QUESTION

A 216π

B 232π

C 432π

D 864π